

JFK AIRPORT

Operations Platform

A cross-enterprise airport operations management system



THE PROBLEM

Airport Operations Are Fragmented

International airport operations must coordinate across multiple independent enterprises, each with unique responsibilities and employees. Without an integrated platform, communication between airlines, airport authorities, and ground service providers is fragmented.

What Breaks When Nobody Talks

01

Delayed Gate Assignments

Flight schedulers and the port authority can't coordinate in real time, so gates get assigned late or double-booked.

02

Missed Ground Handling Windows

Without a shared queue, GroundMaster crews don't know when or where to be, and turnarounds slip.

03

Unresolved Passenger Disruptions

Cancellations, delays, and overbookings bounce between airline staff with no audit trail or clear owner.

04

No Cross-Enterprise Audit Trail

Safety incidents and permit approvals live in separate systems, so compliance has no single source of truth.

No Single Enterprise Can Do It Alone

Every departure depends on all three entities working in smooth coordination — one missed handoff turns into delays, missed connections, and passenger disruptions.

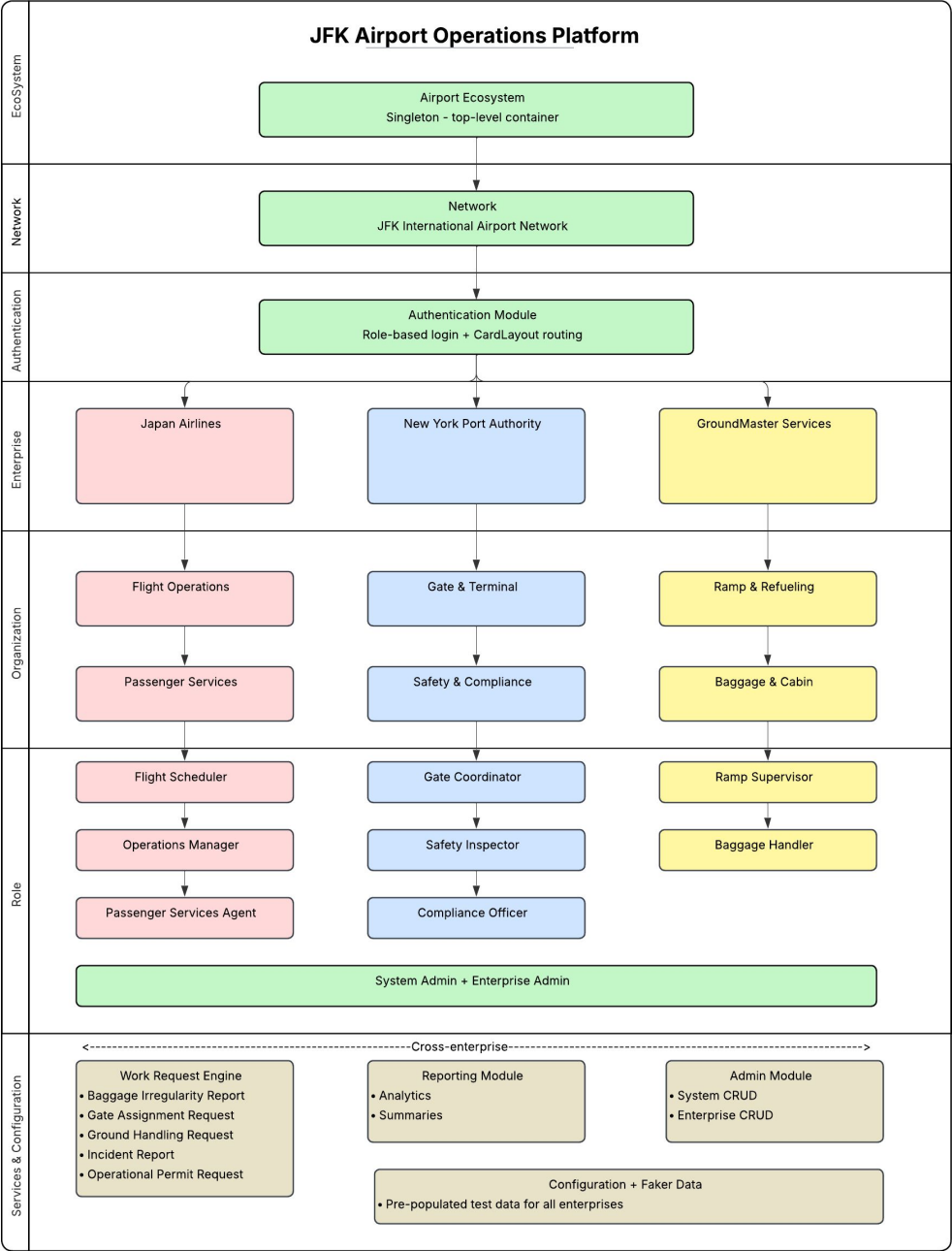


HIGH-LEVEL DIAGRAM

System Architecture

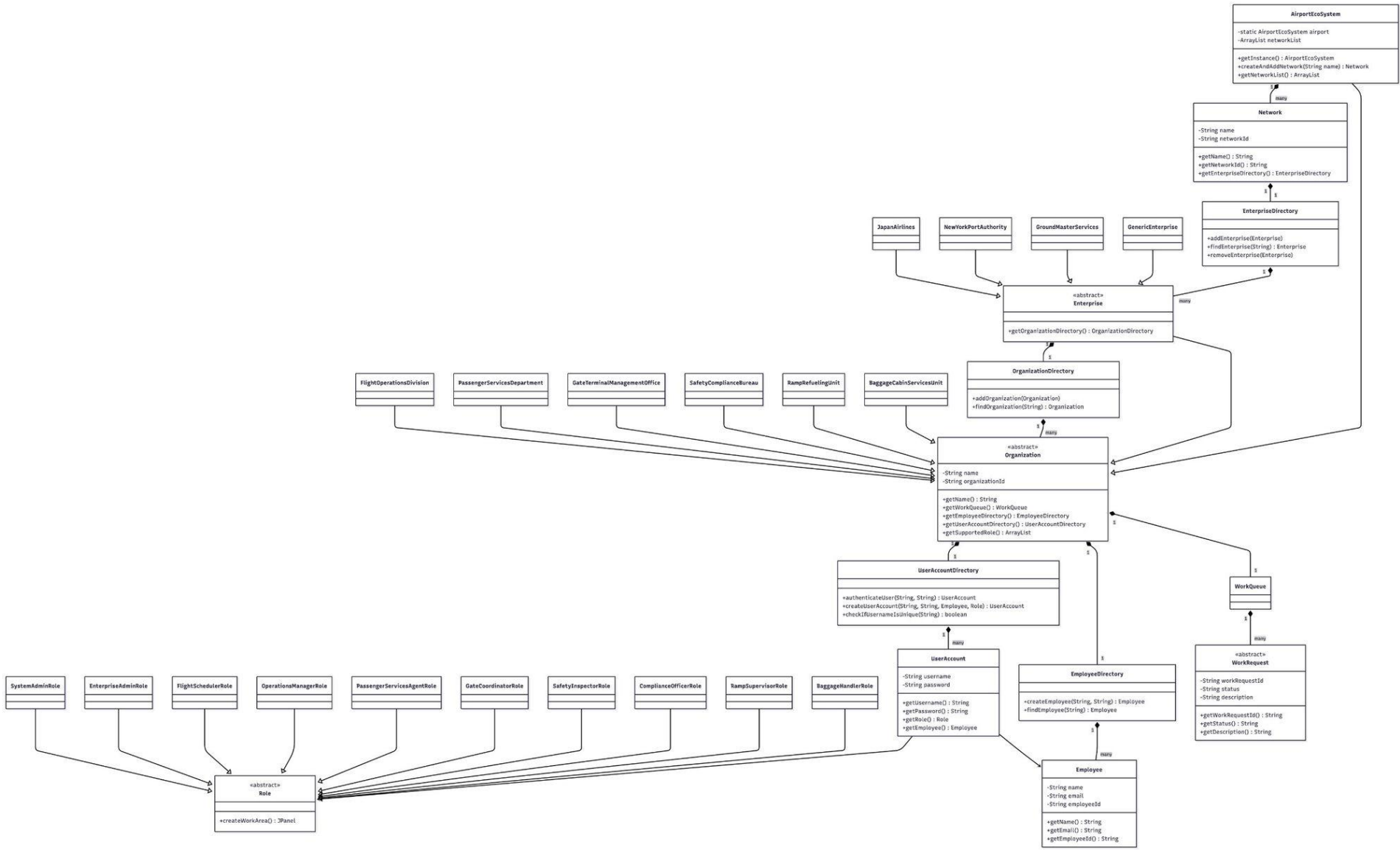
Six layers, top to bottom:

- EcoSystem Singleton container
- Network JFK International
- Authentication Role-based login
- 3 Enterprises JAL · NYPA · GMS
- 6 Organizations Grouped by function
- 10 Roles Plus shared services



Source: JFK Airport High-Level Diagram (team doc)

Complete Class Hierarchy



AirportEcoSystem (Singleton) → Network → 4 Enterprise subclasses → 6 Organizations → 10 Role subclasses → WorkRequest hierarchy

Six Request Types, One Routing Engine

Each request type enforces its own status transitions via `canTransitionTo()`. Cross-enterprise requests land in both sender and receiver queues.

REQUEST TYPE	SENDER	RECEIVER	SCOPE	STATUSES
Gate Assignment Request	Flight Scheduler	Gate Coordinator	Cross-enterprise	Submitted → Under Review → Assigned/Conflict → Confirmed
Ground Handling Request	Flight Scheduler	Ramp Supervisor	Cross-enterprise	Dispatched → Crew Assigned → In Progress → Completed
Operational Permit Request	Operations Manager	Safety Inspector	Cross-enterprise	Submitted → Under Review → Approved / Denied
Baggage Irregularity Report	Baggage Handler	Ramp Supervisor	Cross-org (GMS)	Filed → Under Investigation → Escalated / Resolved
Incident Report & Safety Audit	Safety Inspector	Compliance Officer	Cross-org (NYPA)	Reported → Under Review → Audit Initiated / Closed
Passenger Disruption Request	Passenger Svc Agent	Operations Manager	Cross-org (JAL)	Escalated → Under Review → Authorized / Denied → Resolved

Key Architectural Decisions

Six design choices that make the platform more than a CRUD app.



Singleton Pattern

01

AirportEcoSystem has one guaranteed instance at runtime — no duplicate data, no conflicting authentication states.



Abstract Class Hierarchy

02

Role and Organization are abstract. Every subclass must implement `createWorkArea()` and `getSupportedRole()`.



Role-Based Authentication

03

Login searches across ecosystem, enterprise, and organization levels. Passwords never exposed via a getter.



CardLayout Navigation

04

One JFrame, one window. Card routing switches work areas based on the authenticated user's role.



Cross-Enterprise Routing

05

Work requests are pushed into both sender and receiver queues so both sides see the same live state.



Unique ID Generation

06

Every Employee, UserAccount, Organization, Enterprise, and Network gets an auto-generated ID via a static counter.

Who Built What

PERSON 1

Frederick Triest

- Core class hierarchy and data model
- Role-based authentication system
- System Admin and Enterprise Admin CRUD panels
- ConfigureASystem with pre-populated data
- High-level diagram and project proposal

PERSON 2

Cohen Powell

- Work request engine (all 6 request types)
- Status transition rules via canTransitionTo()
- Cross-enterprise and cross-org routing
- Work area UI panels for all 8 operational roles
- Presentation slides

PERSON 3

Larry Tsao

- Faker data integration across enterprises
- Pre-populated employees and user accounts
- Reporting module with live analytics
- Work request summaries and KPI dashboards
- UML class diagram



Thank You

Questions & Discussion

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